

MATH 324 Winter 2025 - Tutorial 5

Section 003 (Thursday)

February 13, 2025

1 Sufficiency

1.1 Rao-Blackwell theorem

Theorem Suppose $\hat{\theta}$ is unbiased for θ and $\text{Var}(\hat{\theta}) < \infty$. Let U be a sufficient statistic for θ and define $\hat{\theta}^* = E[\hat{\theta}|U]$. Then, for all θ ,

- (1) $E[\hat{\theta}^*] = \theta$, i.e. $\hat{\theta}^*$ is unbiased, and
- (2) $\text{Var}(\hat{\theta}^*) \leq \text{Var}(\hat{\theta})$ for any unbiased estimator of θ .

Thus, $\hat{\theta}^*$ is the MVUE (minimum variance unbiased estimator).

As a result, we have that the MVUE is (1) unbiased and (2) a function of the sufficient statistic.

"Def": Heuristically, the minimal sufficient statistic best summarizes information in the data about θ . Mathematically, for all T sufficient, $U = f(T)$ minimal sufficient.

Question 1 - MVUE for Weibull distribution

Suppose that $Y_1, Y_2, \dots, Y_n$ denote a random sample from the Weibull density function, given by

$$f(y|\theta) = \begin{cases} \frac{2y}{\theta} e^{-y^2/\theta}, & \text{if } y > 0 \\ 0, & \text{elsewhere} \end{cases}$$

Find an MVUE for θ .

$$\begin{aligned} L(y_1, \dots, y_n | \theta) &= \prod_{i=1}^n \frac{2y_i}{\theta} e^{-y_i^2/\theta} \quad (\text{indep}) \\ &= \left(\frac{2}{\theta}\right)^n \left(\prod_{i=1}^n y_i\right) \exp\left\{-\frac{1}{\theta} \sum_{i=1}^n y_i^2\right\} \\ &= \underbrace{\left(\frac{2}{\theta}\right)^n \exp\left\{-\frac{1}{\theta} \sum_{i=1}^n y_i^2\right\}}_{g(\sum_{i=1}^n y_i^2, \theta)} \underbrace{(y_1 y_2 \dots y_n)}_{h(y)} \end{aligned}$$

$\therefore \sum_{i=1}^n Y_i^2$ is sufficient (minimal)
 MVUE will be a function of $U = \sum_{i=1}^n Y_i^2$
 that is unbiased.
 $E[U] = ?$ Let $W = Y_i^2$

$$\begin{aligned} f_W(w) &= f_Y(\sqrt{w}) \frac{d(\sqrt{w})}{dw} = \frac{2\sqrt{w}}{\theta} e^{-w/\theta} \left(\frac{1}{2\sqrt{w}}\right) = \frac{1}{\theta} e^{-w/\theta}, \quad w > 0 \quad \leftarrow \text{exponential p.d.f. with mean } \theta \\ &\Rightarrow E[Y_i^2] = \theta \Rightarrow E[U] = E\left(\sum_{i=1}^n Y_i^2\right) = n\theta \\ &\therefore \hat{\theta} = \frac{1}{n} \sum_{i=1}^n Y_i^2 \text{ is unbiased for } \theta \text{ and} \\ &\quad \text{a function of sufficient stat } U \\ &\therefore \hat{\theta} \text{ is MVUE.} \end{aligned}$$

transformation of a p.d.f.
 $f_U(u) = f_Y(h^{-1}(u)) \frac{dh^{-1}(u)}{du}$
 $U = h(Y)$

Question 2 - Rao-Blackwell Theorem

Let X_1 and X_2 be i.i.d. $N(\theta, 1)$. Consider conditioning the sufficient statistic $\bar{X} = \frac{1}{2}(X_1 + X_2)$ on X_1 , which is not sufficient. Does Rao-Blackwell theorem hold?

$$E(\bar{X}) = \theta, \text{Var}(\bar{X}) = \frac{\sigma^2}{n} = \frac{1}{2}$$

If we condition on X_1 , consider $f(X_1) = E(\bar{X}|X_1)$.

$$E(f(X_1)) = E(E(\bar{X}|X_1)) = E(\bar{X}) = \theta \quad \leftarrow \text{unbiased}$$

~ law of total expectation

$$\text{Var}(f(X_1)) = \text{Var}(E(\bar{X}|X_1)) \leq \text{Var}(\bar{X})$$

law of total variance:

$$\text{Var}(Y) = \underbrace{\text{Var}(E(Y|X))}_{>0} + \underbrace{E(\text{Var}(Y|X))}_{>0}$$

$$f(X_1) = E(\bar{X}|X_1) = E\left(\frac{1}{2}(X_1 + X_2)|X_1\right)$$

$$= \frac{1}{2} E(X_1|X_1) + \frac{1}{2} E(X_2|X_1) \quad \leftarrow X_2 \text{ indep. of } X_1, E(X_2|X_1) = E(X_2)$$

$$= \frac{1}{2} X_1 + \frac{1}{2} \theta \quad \text{Not an estimator (still depends on } \theta)$$

$\therefore f(X_1)$ is not an estimator $\Rightarrow$ R-B does not hold (since X_1 is not sufficient)

Question 3 - MVUE for Poisson random variable

(Exercise 9.59) The number of breakdowns Y per day for a certain machine is a Poisson random variable with mean λ . The daily cost of repairing these breakdowns is given by $C = 3Y^2$. If $Y_1, Y_2, \dots, Y_n$ denote the observed number of breakdowns for n independently selected days, find an MVUE for $E[C]$.

$$Y \sim \text{Pois}(\lambda), E[C] = E[3Y^2] = 3E[Y^2] = 3(\text{Var}(Y) + (E(Y))^2) = 3(\lambda + \lambda^2)$$

We know that $\sum_{i=1}^n Y_i$ is sufficient for λ (Exer 9.39)

Looking for $\hat{\theta}$ unbiased for $E[C]$ and fcn of $\sum_{i=1}^n Y_i \Rightarrow$ by Rao-Blackwell, $\hat{\theta}$ is MVUE

$$\sum_{i=1}^n Y_i \sim \text{Poisson}(n\lambda) \Rightarrow E(\bar{Y}) = \text{Var}(\bar{Y}) + (E(\bar{Y}))^2 = \frac{\lambda}{n} + \lambda^2$$

$$\text{As well, } E\left(\frac{\bar{Y}}{n}\right) = \frac{\lambda}{n}$$

$$\lambda^2 = E(\bar{Y}) - E\left(\frac{\bar{Y}}{n}\right) \Rightarrow 3\left[\bar{Y}^2 - \frac{\bar{Y}}{n} + \bar{Y}\right] = 3\left[\bar{Y}^2 + \bar{Y}\left(1 - \frac{1}{n}\right)\right] = \hat{\theta}$$

$\rightarrow$ unbiased

$\rightarrow$ function of sufficient stat.

$\Rightarrow \hat{\theta}$ is MVUE for $E[C]$.

Question 4 - MVUE using Rao-Blackwell

(Exercise 9.51/9.62) Let $Y_1, Y_2, \dots, Y_n$ denote a random sample from the probability density function

$$f(y|\theta) = \begin{cases} e^{-(y-\theta)}, & y \geq \theta \\ 0, & \text{elsewhere} \end{cases}$$

(a) Show that $Y_{(1)} = \min(Y_1, \dots, Y_n)$ is sufficient for θ .

$$f_{Y_{(1)}} = n[1-F(y)]^{n-1}f(y) \quad (\text{from Sec 6.7})$$

$$1-F(y) = e^{-(y-\theta)}$$

(b) Find a function of $Y_{(1)}$ that is an MVUE for θ .

(a) Using the indicator function, we can write

$$f(y|\theta) = e^{-(y-\theta)} I_{(\theta, \infty)}(y)$$

$$\text{Then, } L(\theta) = \prod_{i=1}^n e^{-(y_i-\theta)} I_{(\theta, \infty)}(y_i) \quad (\text{indep})$$

$$= \exp\left\{\sum_{i=1}^n y_i + n\theta\right\} \prod_{i=1}^n I_{(\theta, \infty)}(y_i)$$

$$= \underbrace{\exp\left\{\sum_{i=1}^n y_i\right\}}_{h(y)} \underbrace{e^{n\theta} I_{(\theta, \infty)}(y_{(1)})}_{g(y_{(1)}, \theta)}$$

$\therefore Y_{(1)}$ is sufficient by Thm 9.4

only nonzero if the smallest of the y_i 's is $> \theta$

$$(b) E(Y_{(1)}) = \int_{\theta}^{\infty} y f_{Y_{(1)}}(y) dy$$

$$= \int_{\theta}^{\infty} n y e^{-n(y-\theta)} dy$$

$$\text{let } u = y - \theta$$

$$= \int_0^{\infty} n(u+\theta) e^{-nu} du$$

$$= \theta + \frac{1}{n}$$

$\Rightarrow Y_{(1)} - \frac{1}{n}$ is unbiased and a function of a sufficient statistic

$\therefore \hat{\theta} = Y_{(1)} - \frac{1}{n}$ is MVUE for θ

Question 5 - MVUE and its variance

(Exercise 9.64) Let $Y_1, \dots, Y_n$ be a random sample from a normal distribution with mean μ and variance 1.

(a) Show that MVUE of μ^2 is $\hat{\mu}^2 = \bar{Y}^2 - 1/n$.

(b) Derive the variance of $\hat{\mu}^2$.

(a) We know $\bar{Y}$ is sufficient for μ in the Normal distribution.

$$\text{Since } \sigma^2 = 1, \bar{Y} \sim N(\mu, \frac{1}{n}) \Rightarrow E[\bar{Y}^2] = \text{Var}(\bar{Y}) + (E[\bar{Y}])^2 = \frac{1}{n} + \mu^2$$

$\therefore \hat{\mu}^2 = \bar{Y}^2 - \frac{1}{n}$ is unbiased and a function of the sufficient $\bar{Y}$

By Rao-Blackwell, it is the MVUE.

$$(b) \text{Var}(\hat{\mu}^2) = \text{Var}(\bar{Y}^2 - \frac{1}{n}) = \text{Var}(\bar{Y}^2) = E[\bar{Y}^4] - \underbrace{(E[\bar{Y}^2])^2}_{= \frac{1}{n} + \mu^2}$$

Scalar with 0 variance

$$\text{Use the MGF of Normal to get } E[\bar{Y}^4] = \frac{3}{n^2} + \frac{6\mu^2}{n} + \mu^4$$

$$M(t) = \exp(\mu t + \frac{t^2 \sigma^2}{2})$$

$$E[X^4] = M^{(4)}(0)$$

$$\Rightarrow \text{Var}(\hat{\mu}^2) = \frac{3}{n^2} + \frac{6\mu^2}{n} + \mu^4 - (\frac{1}{n} + \mu^2)^2$$

$$= \frac{2 + 4n\mu^2}{n^2}$$

Question 6 - Rao-Blackwell application for Poisson

(Exercise 9.106) Suppose that $Y_1, Y_2, \dots, Y_n$ denote a random sample from a Poisson distribution with mean λ . Find the MVUE of $P(Y_i = 0) = e^{-\lambda}$. (Hint: Construct a binary random variable with expectation $E[T] = e^{-\lambda}$.)

Let $T=1$ if $Y_1=0$ and $T=0$ otherwise. $\Rightarrow E(T) = 1 \cdot P(T=1) = P(Y_1=0) = e^{-\lambda}$

$W = \sum_{i=1}^n Y_i \sim \text{Poisson}(n\lambda)$. W is sufficient for λ .

$$E(T|W=w) = P(T=1|W=w) = P(Y_1=0|W=w) = \frac{P(Y_1=0, W=w)}{P(W=w)}$$

$$= \frac{P(Y_1=0) P(\sum_{i=2}^n Y_i=w)}{P(W=w)} \quad \text{b/c if } Y_1=0$$

$$\Rightarrow \sum_{i=1}^n Y_i = \sum_{i=2}^n Y_i \sim \text{Poisson}((n-1)\lambda)$$

$$= \frac{e^{-\lambda} \left[e^{-(n-1)\lambda} \frac{[(n-1)\lambda]^w}{w!} \right]}{e^{-n\lambda} \frac{(n\lambda)^w}{w!}} = \left(\frac{n-1}{n} \right)^w = \left(1 - \frac{1}{n} \right)^w$$

$\therefore$ MVUE is $\left(1 - \frac{1}{n} \right)^{\sum_{i=1}^n Y_i}$